United States Patent	12417436
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Hajian; Hamid

Automated parameterized modeling and scoring intelligence system

Abstract

An automated parameterized modeling and scoring intelligence system includes a parameterized score estimation software tool and a parameterized score optimization software tool. The parameterized score estimation software tool processes design metrics associated with a current project according to historical project data selected based on a similarity with at least some of the design metrics to determine a score estimation for the current project. The parameterized score optimization software tool processes the score estimation based on external application data retrieved from an external application to determine an expected yield for the current project. A user of the system may iterate against the score estimation or the expected yield by changing one or more of the parameters used to determine same. The iteration may result in a score estimation or expected yield different from the initial versions thereof, such as to identify an optimal design for the current project.

Inventors:	**Hajian; Hamid (Walnut Creek, CA)**
Applicant:	**Zebel Group, Inc.** (Walnut Creek, CA)
Family ID:	**1000008819246**
Assignee:	**Zebel Group, Inc. (Walnut Creek, CA)**
Appl. No.:	**18/147328**
Filed:	**December 28, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230128318 A1	Apr. 27, 2023

Related U.S. Application Data

continuation parent-doc US 16676876 20191107 US 11568367 child-doc US 18147328

Publication Classification

Int. Cl.: **G06Q10/10** (20230101); **G06F16/951** (20190101); **G06F17/18** (20060101); **G06N20/00** (20190101); G06F8/10 (20180101)

U.S. Cl.:

CPC **G06Q10/103** (20130101); **G06F16/951** (20190101); **G06F17/18** (20130101); **G06N20/00** (20190101); G06F8/10 (20130101)

Field of Classification Search

CPC: G06Q (10/103); G06N (20/00); G06F (17/18); G06F (8/10); G06F (8/20)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2005/0234761	12/2004	Pinto	705/7.31	G06Q 10/0635
2006/0117317	12/2005	Crawford	718/104	G06Q 10/0631
2006/0178918	12/2005	Mikurak	705/7.31	G06Q 10/06
2008/0313596	12/2007	Kreamer	717/101	G06Q 10/06315
2008/0319811	12/2007	Casey	705/7.13	G06Q 10/06311
2010/0131314	12/2009	Lo Yuk Ting	705/7.11	G06Q 10/063
2013/0159242	12/2012	Rahmouni	706/52	G06Q 10/06313
2015/0294246	12/2014	Guven Kaya	705/7.28	G06Q 10/06393
2016/0342690	12/2015	Young	N/A	G06Q 50/20
2018/0060759	12/2017	Chu	N/A	G06N 20/00
2019/0205484	12/2018	Morkos	N/A	G06Q 10/06313

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2019537125	12/2015	JP	G06Q 10/10

OTHER PUBLICATIONS

English translation of JP2019537125A (Year: 2016). cited by examiner

J. J. Cuadrado-Gallego, P. Rodríguez-Soria and B. Martín-Herrera, “Analogies and Differences between Machine Learning and Expert Based Software Project Effort Estimation,” 2010 11th ACIS International Conference on Software Engineering (Year: 2010). cited by examiner

Cuadrado-Gallego et al., “Analogies and Differences between Machine Leaming and Expert Based

Software Project Effort Estimation," 11th ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed, 2010, pp. 269-276. cited by applicant

---

Primary Examiner: O'Connor; Jerry

Assistant Examiner: Gunn; Jeremy L

Attorney, Agent or Firm: Young Basile Hanlon & MacFarlane, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
(1) This application claims priority to and the benefit of U.S. patent application Ser. No. 16/676,876, filed Nov. 7, 2019, the entire disclosure of which is hereby incorporated by reference.
TECHNICAL FIELD
(2) This disclosure relates to an automated parameterized modeling and scoring intelligence system.
BACKGROUND
(3) Project design planning typically incurs significant time and manual user effort to tediously prepare, revise, and analyze designs. A typical approach may include drafting an initial design according to some limited set of initial design parameters, submitting the designs for review, awaiting design feedback, revising the designs according to the feedback, and potentially suffering late-stage design changes due to previously unconsidered options. Some solutions exist for improving the project design process, but they suffer from several drawings. For example, conventional solutions require detailed, specific parameter information, which may not be available until a later stage of project development. In another example, conventional solutions also fail to leverage highly valuable historical project data such as by recognizing precise similarities between historical projects and a new project for which a design is prepared.
(4) The technical implementation of conventional solutions also suffers from drawbacks in that the platforms are not intelligent, not automated, and not extensible. For example, such conventional solutions do not enable users to extend parameters or change parameters during processing. In another example, such conventional solutions do not automate parameter modeling and scoring, such as because manual user intervention is required at each step and/or because such conventional solutions lack intelligence functionality which analyzes, maps, and/or scores parameterized information. In yet another example, such conventional solutions are not implemented in technical service environments which can be represented using client/server functionality, single or multi-tenant computing environments, or otherwise.
SUMMARY
(5) Disclosed herein are, inter alia, implementations of systems and techniques for automated parameterized modeling and scoring intelligence.
(6) In one implementation, a system for automated parameterized modeling and scoring intelligence comprises a server device included in a service environment and a database. The server device includes a processor, a memory, and a network interface. The network interface enables communications between the server device and a client device. The memory stores instructions executable by the processor for implementing functionality of an automated parameterized modeling and scoring intelligence platform. The database stores information associated with one or more users of the automated parameterized modeling and scoring intelligence platform. The information includes one or more sets of historical project data. The instructions include

instructions to: define parameters representing design metrics associated with a current project based on first input received from the client device; select one or more historical projects for modeling the first input parameters, wherein a set of historical project data of the sets of historical project data corresponds to each of the one or more historical projects; determine a score estimation for the current project based on the defined parameters and according to the set of historical project data; perform a regression analysis against the score estimation based on external application data to determine value estimate data for the current project, wherein the external application data is associated with an external application implemented in an external environment separate from the service environment; generate a pro forma for the current project based on the value estimate data; identify one or more first expected yield candidates for the current project based on feedback responsive to the pro forma received from the client device; change one or more of the defined parameters, the set of historical project data, or the external application data in response to a request received from the client device; identify one or more second expected yield candidates for the current project based on the changed one or more of the defined parameters, the set of historical project data, or the external application data; receive a selection of one of the one or more second expected yield candidates as an expected yield; and output a project design corresponding to the expected yield to the client device.

(7) In another implementation, a system for automated parameterized modeling and scoring intelligence comprises a server device that implements an automated parameterized modeling and scoring intelligence platform and a database that stores information associated with one or more users of the automated parameterized modeling and scoring intelligence platform. The automated parameterized modeling and scoring intelligence platform includes a parameterized score estimation software tool and a parameterized score optimization software tool. The parameterized score estimation software tool processes design metrics associated with a current project according to historical project data selected based on a similarity with at least some of the design metrics to determine a score estimation for the current project. The parameterized score optimization software tool processes the score estimation for the current project based on external application data retrieved from an external application to determine an expected yield for the current project and to output a design associated with the expected yield for the current project.

(8) In yet another implementation, a system for automated parameterized modeling and scoring intelligence comprises a server device including a processor and a memory, wherein the processor executes instructions stored in the memory. The instructions include instructions to: receive, from a client device, design metrics associated with a current project and a selection of one or more historical projects; determine a score estimation for the current project based on the design metrics and according to historical project data associated with the one or more historical projects; receive, from an external application, external application data associated with one or more real property sites; determine value estimate data for the current project by performing a regression analysis against the score estimation based on the external application data; identify a set of expected yield candidates based on the value estimate data, wherein each expected yield candidate of the set of expected yield candidates is associated with a design for the current project; receive, from the client device, a selection of one of the expected yield candidates of the set of expected yield candidates; and transmit, to the client device, the design associated with the selected expected yield candidate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various

features are arbitrarily expanded or reduced for clarity.
(2) FIG. **1** is a block diagram showing an example of an automated parameterized modeling and scoring intelligence system.
(3) FIG. **2** is a block diagram showing an example of an automated parameterized modeling and scoring intelligence platform of an automated parameterized modeling and scoring intelligence system.
(4) FIG. **3** is a block diagram showing examples of software tools of an automated parameterized modeling and scoring intelligence platform.
(5) FIG. **4** is a block diagram of examples of functionality of a parameterized score estimation software tool of an automated parameterized modeling and scoring intelligence platform.
(6) FIG. **5** is a block diagram of examples of functionality of a parameterized score optimization software tool of an automated parameterized modeling and scoring intelligence platform.
(7) FIG. **6** is an illustration showing an example of a graphical user interface (GUI) of a parameterized score estimation software tool.
(8) FIG. **7** is an illustration showing an example of a GUI of a parameterized score optimization software tool.
(9) FIG. **8** is a flowchart showing an example of a technique for parameterized score estimation using an automated parameterized modeling and scoring intelligence system.
(10) FIG. **9** is a flowchart showing an example of a technique for parameterized score optimization using an automated parameterized modeling and scoring intelligence system.
(11) FIG. **10** is a block diagram showing an example of a computing device which may be used in an automated parameterized modeling and scoring intelligence system.

DETAILED DESCRIPTION

(12) Reference is first made to examples of hardware and software structures used to implement an automated parameterized modeling and scoring intelligence system. FIG. **1** is a block diagram showing an example of an automated parameterized modeling and scoring intelligence system **100**. The automated parameterized modeling and scoring intelligence system **100** can be or include a distributed computing system (e.g., a client-server computing system), a cloud computing system, a clustered computing system, or the like.
(13) The automated parameterized modeling and scoring intelligence system **100** includes a user environment **102**, which may be or otherwise refer to devices, configurations, objects, or the like under the possession, control, or use by a public entity, private entity, or other corporate entity or individual that purchases or otherwise uses software services. The user environment **102** includes one or more clients. For example, and without limitation, the user environment **102** can include a client **104**. The client **104** can include one or more computing devices, such as a mobile phone, a tablet computer, a laptop computer, a notebook computer, a desktop computer, or another suitable computing device or combination of computing devices.
(14) In some implementations, the client **104** can be implemented as a single physical unit or as a combination of physical units. In some implementations, a single physical unit can include multiple clients. In some implementations, the client **104** can be an instance of software running on a device associated with the user environment **102**. As used herein, the term “software” can include, but is not limited to, applications, programs, instances, processes, threads, services, plugins, patches, application version upgrades, or another identifiable computing aspect.
(15) A network **106** connects the user environment **102** and a service environment **108**. The service environment **108** can include one or more servers used implement software, such as Platform-as-a-Service (PaaS) software, Software-as-a-Service (SaaS) software, software specific to mobile devices, web application software, or other software. For example, and without limitation, the service environment **108**, as generally illustrated, includes an application server **112** which implements an application node **114** and a database server **116** which implements a database **118**.
(16) In some implementations, the servers in the service environment **108** may be located at a

datacenter. A datacenter can represent a geographic location, which can include a facility, where the one or more servers are located. The automated parameterized modeling and scoring intelligence system **100** can include a number of datacenters and servers, or it can include a configuration of datacenters and servers different from that generally illustrated in FIG. **1**. In some implementations, the service environment **108** can be associated or communicate with one or more datacenter networks or domains, which can include domains other than a client domain, such as which can be associated with the client **104**.

(17) The client **104** and the servers associated with the service environment **108** may be configured to connect to, or communicate via, the network **106**. Furthermore, the client **104** associated with the user environment **102** can connect to the network **106** via a communal connection point, link, or path or using a distinct connection point, link, or path. A connection point, link, or path can be wired, wireless, use other communications technologies, or a combination thereof.

(18) The network **106** can be or include the Internet. Alternatively, the network **106** can be or include a local area network (LAN), a wide area network (WAN), a virtual private network (VPN), or another public or private means of electronic computer communication capable of transferring data between a client, such as the client **104**, and one or more servers associated with the service environment **108**, or a combination thereof. The network **106**, the service environment **108**, or any other element, or combination of elements, of the automated parameterized modeling and scoring intelligence system **100** can include network hardware such as routers, switches, load balancers, other network devices, or combinations thereof. For example, the service environment **108** can include a load balancer **110** for routing traffic from the network **106** to various servers associated with the service environment **108**.

(19) The load balancer **110** can route, or direct, computing communications traffic, such as signals or messages, to respective elements of the service environment **108**. For example, the load balancer **110** can operate as a proxy, or reverse proxy, for a service, such as an Internet-delivered service, provided by the service environment **108** to one or more remote clients, such as the client **104**, via the network **106**. Routing functions of the load balancer **110** can be configured directly or via a Domain Name System (DNS) service. The load balancer **110** can coordinate requests from remote clients, such as the client **104**, and can simplify client access by masking the internal configuration of the service environment **108** from the remote clients. The load balancer **110** can operate as a firewall, allowing or preventing communications based on configuration settings.

(20) Although the load balancer **110** is depicted in FIG. **1** as being within the service environment **108**, in some implementations, the load balancer **110** can instead be located outside of the service environment **108**, for example, when providing global routing for multiple datacenters. In some implementations, load balancers can be included both within and outside of the service environment **108**. The inclusion of the load balancer **110** within the service environment **108** does not itself represent that the load balancer **110** is under the direct or indirect possession and/or control of a software provider who provides software implemented using a server of the service environment **108**.

(21) In the example shown and as described above, the service environment **108** includes the application server **112** and the database server **116**. The application server **112** and/or the database server **116** can be a computing system, which can include one or more computing devices, such as a desktop computer, a server computer, or any other computer capable of operating as a server. In some implementations, the application server **112** and/or the database server **116** can be non-hardware servers implemented on a physical device, such as a hardware server. In some implementations, the application server **112** and the database server **116** can be implemented as a single hardware server or as a single non-hardware server implemented on a single hardware server. In some implementations, the service environment **108** can include servers other than or in addition to the application server **112** or the database server **116**, for example, a web server.

(22) The application node **114** is or otherwise refers to a process executed on the application server

112. For example, and without limitation, the application node **114** can be executed in order to deliver services to a client, such as the client **104**, as part of a web application, either as standalone software or as an online component to other software. The application node **114** can be implemented using processing threads, virtual machine instantiations, or other computing features of the application server **112**. In some implementations, the application node **114** can store, evaluate, or retrieve data from a database, such as the database **118** of the database server **116**.

(23) The application server **112** can include a number of application nodes, depending upon a system load or other characteristics associated with the application server **112**. For example, and without limitation, the application server **112** can include two or more nodes forming a node cluster. The application nodes implemented on a single application server **112** may run on different hardware servers.

(24) The database server **116** can be configured to store, manage, or otherwise provide data for delivering services to the client **104** over a network. The database server **116** may include a data storage unit, such as the database **118**, which can be accessible by an application executed on the application node **114**. The database **118** may be implemented as a relational database, an object database, an XML database, a configuration management database (CMDB), a management information base (MIB), one or more flat files, other suitable non-transient storage mechanisms, or a combination thereof. While limited examples are described, the database **118** can be configured as or comprise another suitable database type. Further, the automated parameterized modeling and scoring intelligence system **100** can include one, two, three, or another number of databases configured as or comprising another database type or combination thereof.

(25) In some implementations, one or more databases (e.g., the database **118**), tables, other information sources, or portions or combinations thereof may be stored, managed, or otherwise provided by one or more of the elements of the automated parameterized modeling and scoring intelligence system **100** other than the database server **116**, such as the client **104** or the application server **112**.

(26) Some or all of the systems and techniques disclosed herein can operate or be executed on or by the servers associated with the automated parameterized modeling and scoring intelligence system **100**. In some implementations, the systems and techniques disclosed, portions thereof, or combinations thereof can be implemented on a single device, such as a single server, or a combination of devices, for example, a combination of the client **104** and a server which implements the application server **112** and/or the database server **116**.

(27) In some implementations, the automated parameterized modeling and scoring intelligence system **100** can include devices other than the client **104**, the load balancer **110**, and a server which implements the application server **112** and/or the database server **116**. In some implementations, one or more additional servers can operate as system infrastructure control aspects, from which servers, clients, or both servers and clients, can be monitored, controlled, configured, or a combination thereof.

(28) The service environment **116** can allocate resources of the user environment **102** using a single-tenant architecture or a multi-tenant architecture. Allocating resources in a multi-tenant architecture can include installations or instantiations of one or more servers, such as application servers, database servers, or another server or a combination of servers, which can be shared amongst multiple user environments or multiple accounts within a single user environment. For example, a web server, such as a unitary Apache installation; an application server, such as a unitary Java Virtual Machine (JVM); or a single database server catalog, such as a unitary MySQL catalog, can handle requests from multiple user environments or multiple accounts within a single user environment. In some implementations of a multi-tenant architecture, the application server, the database server, or both can distinguish between and segregate data or other information of the various users of the system.

(29) In a single-tenant infrastructure, separate web servers, application servers, database servers, or

combinations thereof can be provisioned individually for at least some user environments or accounts within a user environment. User environments or accounts within a user environment can access one or more dedicated web servers, have transactions processed using one or more dedicated application servers, or have data stored in one or more dedicated database servers, catalogs, or both. Physical hardware servers can be shared such that multiple installations or instantiations of web servers, application servers, database servers, or combinations thereof can be installed on the same physical server. An installation can be allocated a portion of the physical server resources, such as memory, storage, communications bandwidth, or processor cycles.
(30) A user environment instance can include multiple web server instances, multiple application server instances, multiple database server instances, or a combination thereof. The server instances can be physically located on different physical servers and can share resources of the different physical servers with other server instances associated with other customer instances. In a distributed computing system, multiple user environment instances can be used concurrently. Other configurations or implementations of user environment instances can also be used. The use of user environment instances in a single-tenant architecture can provide true data isolation from other user environment instances, amongst other benefits.
(31) FIG. **2** is a block diagram showing an example of an automated parameterized modeling and scoring intelligence platform **200** of an automated parameterized modeling and scoring intelligence system, such as the automated parameterized modeling and scoring intelligence system **100** shown in FIG. **1**. The automated parameterized modeling and scoring intelligence platform **200** is software that receives and processes requests **202** to determine scores **204**. The automated parameterized modeling and scoring intelligence platform **200** is implemented within a service environment **206**, which may, for example, be the service environment **108** shown in FIG. **1**. The service environment **206** includes an internal database **204** which is accessed by the automated parameterized modeling and scoring intelligence platform **200** to process the requests **202** and determine the scores **204**. The internal database **204** may, for example, be the database **118** shown in FIG. **1**.
(32) The requests **202** are requests for parametric modeling of data using automated and intelligence functionality of the automated parameterized modeling and scoring intelligence platform **200**. A request may include information used as input to the automated parameterized modeling and scoring intelligence platform **200**. The scores **204** are scores determined using parametric modeling of data associated with the requests **202**, such as by using the automated and intelligence functionality of the automated parameterized modeling and scoring intelligence platform **200**. A score may be or refer to information output from the automated parameterized modeling and scoring intelligence platform **200**.
(33) For example, in implementations in which the automated parameterized modeling and scoring intelligence platform **200** is or includes software for parameterized score estimation for construction projects, a request may include or indicate a set of design metrics representing first input parameter values for a current project and a set of historical project data representing second input parameter values. A score resulting from the processing of the request may include or indicate a total estimated value attributed to the completion of the construct project based on the set of design metrics and according to the set of historical project data.
(34) In another example, in implementations in which the automated parameterized modeling and scoring intelligence platform **200** is or includes software for parameterized score optimization for construction projects, a request may include or indicate a set of design metrics representing first input parameter values for a current project, a set of historical project data representing second input parameter values, and external application data corresponding to real-world property information, such as market value attributions, representing third input parameter values. A score resulting from the processing of the request may include or indicate an optimized score value attributed to the overall yield of the construct project according to the set of design metrics and based on the set of historical project data and the external application data.

(35) In some implementations, the automated parameterized modeling and scoring intelligence platform **200** may be a platform of a PaaS software offering. For example, the automated parameterized modeling and scoring intelligence platform **200** may refer to a single-tenant platform for automated parameterized modeling and scoring intelligence. In such a case, the information within the internal database **208** which is used by the automated parameterized modeling and scoring intelligence platform **200** may be specific to the user environment or account for which the automated parameterized modeling and scoring intelligence platform **200** is instantiated. In another example, the automated parameterized modeling and scoring intelligence platform **200** may refer to a multi-tenant platform for automated parameterized modeling and scoring intelligence. In such a case, the information within the internal database **208** which is used by the automated parameterized modeling and scoring intelligence platform **200** may correspond to multiple user environments or multiple accounts for which the automated parameterized modeling and scoring intelligence platform **200** is instantiated.

(36) In some implementations, the automated parameterized modeling and scoring intelligence platform **200** may be some or all of a SaaS software offering. For example, the automated parameterized modeling and scoring intelligence platform **200** may refer to online or offline software for automated parameterized modeling and scoring intelligence. Similar to with implementations of a PaaS software offering, the SaaS software offering may also include or otherwise leverage a single-tenant structure or a multi-tenant structure.

(37) The internal database **208** is a database included within or otherwise which is accessible by computing aspects of the service environment **206**. The internal database **208** stores data used by the automated parameterized modeling and scoring intelligence platform **200**. As such, the internal database **208** may be considered a live database of information for aspects processed by or otherwise using the automated parameterized modeling and scoring intelligence platform **200**. For example, the data stored within the internal database **208** may include data used by a parameterized score estimation software tool of the automated parameterized modeling and scoring intelligence platform **200** and/or by a parameterized score optimization software tool of the automated parameterized modeling and scoring intelligence platform **200**. In the example in which the automated parameterized modeling and scoring intelligence platform **200** is or includes software for parameterized score estimation and optimization for construction projects, the data stored within the internal database **208** may be or include historical project data associated with one or more previous construction projects.

(38) The internal database **208** may be structured based on the particular tenancy-based implementation of the automated parameterized modeling and scoring intelligence platform **200**. For example, where the automated parameterized modeling and scoring intelligence platform **200** is single-tenant, the internal database **208** may include an entire database storing information dedicated to the user environment or account for which the single-tenant instance is instantiated. In another example, where the automated parameterized modeling and scoring intelligence platform **200** is multi-tenant, the internal database **208** may include a number of database tables each storing records corresponding to individual user environments or accounts.

(39) In some implementations, the use of the automated parameterized modeling and scoring intelligence platform **200** may first include initializing the internal database **208**. Initializing the internal database **208** may include a user of the automated parameterized modeling and scoring intelligence platform **200** uploading or otherwise storing, or causing to be stored, sets of data which may be processed using one or more software tools of the automated parameterized modeling and scoring intelligence platform **200**. For example, in the example in which the automated parameterized modeling and scoring intelligence platform **200** is or includes software for parameterized score estimation for construction projects, the user of the automated parameterized modeling and scoring intelligence platform **200** may upload design metrics, historical project data, and/or other information which may be processed to estimate and/or optimize the scores **204**. The

uploading or other storing, or causing to be stored, of such data within the internal database **208** represents a one-time action such that the same data does not need to be stored in the internal database **208** again thereafter.

(40) In some implementations, the automated parameterized modeling and scoring intelligence platform **200** may be configured to communicate with computing aspects within one or more external environments, such as an external environment **210**. The external environment **210** may include an external application **212** representing software functionality implemented outside of the service environment **206**. The external environment **210** may also include an external database **214** storing data used by or otherwise to implement the external application **212**. The automated parameterized modeling and scoring intelligence platform **200** may be configured to receive information stored within the external database **214** such as through the external application **212**. For example, an application programming interface (API) of the external application **212** may be made available to the automated parameterized modeling and scoring intelligence platform **200**. An API call to the external application **212** may be made from the automated parameterized modeling and scoring intelligence platform **200**, such as to retrieve certain data from within the external database **214** or to otherwise cause the external application **212** to transmit such data to the automated parameterized modeling and scoring intelligence platform **200**.

(41) For example, in implementations in which the automated parameterized modeling and scoring intelligence platform **200** is or includes software for parameterized score optimization for construction projects, the external database **214** can store external application data corresponding to real-world property information, such as market value attributions, used as input to the automated parameterized modeling and scoring intelligence platform **200**.

(42) The software tools of the automated parameterized modeling and scoring intelligence platform **200** may be configurable, such as by enabling a user of the automated parameterized modeling and scoring intelligence platform **200** to modify inputs to those software tools based on reported outputs from those software tools. For example, in implementations in which the automated parameterized modeling and scoring intelligence platform **200** is or includes software for parameterized score estimation for construction projects, a score representing a relatively high total estimated value may be reported to a user, who may then reconfigure values one or more of the design metrics used as input to the automated parameterized modeling and scoring intelligence platform **200**, such as to achieve a different score thereby.

(43) FIG. **3** is a block diagram showing an example of software tools, interfaces, and models of an automated parameterized modeling and scoring intelligence platform **300**, such as the automated parameterized modeling and scoring intelligence platform **200** shown in FIG. **2**. In the example as generally shown, the automated parameterized modeling and scoring intelligence platform **300** includes a parameterized score estimation software tool **302**, a parameterized score optimization software tool, a web crawler deployment software tool **306**, an application and database interfaces **308**, GUIs **310**, and machine learning models **312**.

(44) The parameterized score estimation software tool **302** is a software tool which processes a limited set of inputs to determine and output a score estimation. The parameterized score estimation software tool **302** extrapolates upon the limited set of inputs according to one or more selected processing criteria to determine the score estimation. In the example in which the parameterized score estimation software tool **302** is used for estimating scores for construction projects, the limited set of inputs may be or otherwise refer to a set of design metrics representing first input parameter values, and the one or more selected processing criteria may be or otherwise refer to a set of historical project data representing second input parameter values.

(45) The design metrics may include values received as input from a user of the parameterized score estimation software tool **302**. Alternatively, the design metrics may include values extracted automatically by the parameterized score estimation software tool **302**, such as from a set of user data. The set of user data may refer to information stored within a database, such as a database of

the automated parameterized modeling and scoring intelligence platform **300** (e.g., the internal database **208** shown in FIG. **2**) or a database in an environment external to the automated parameterized modeling and scoring intelligence platform **300** (e.g., the external database **314** shown in FIG. **2**). As a further alternative, the design metrics may include values generated using intelligence aspects of the automated parameterized modeling and scoring intelligence platform **300**, for example, one or more of the machine learning models **312**.

(46) The one or more selected processing criteria may include indications of historical projects, which are constructions projects which have already been completed or otherwise for which design considerations have been finalized. A historical project may be represented by a set of historical project data, which may be stored in a database of the automated parameterized modeling and scoring intelligence platform **300** (e.g., the internal database **208**) or a database in an environment external to the automated parameterized modeling and scoring intelligence platform **300** (e.g., the external database **314**). The user of the parameterized score estimation software tool **302** may select one or more historical projects to use as guide criteria for the score estimation determination. Alternatively, the one or more historical projects may be selected by the parameterized score estimation software tool **302** or another aspect of the automated parameterized modeling and scoring intelligence platform **300**, for example, using one or more of the machine learning models **312**.

(47) The score estimation determined and output by the parameterized score estimation software tool **302** may include or indicate a total estimated value attributed to the completion of the current project based on the limited set of inputs and according to the one or more selected processing criteria. The total estimated value may represent or otherwise indicate an estimated amount of one or more resources to use to complete the current project.

(48) After a first score estimation is output, a user of the parameterized score estimation software tool **302** may modify one or more values of the design metrics, enable or disable one or more design metric categories, and/or change the selection of historical projects used. After making such changes, the user may cause the parameterized score estimation software tool **302** to determine a new score based on the changes. The score estimation output after the changes are made may not be the same as the score estimation output before the changes are made. In some implementations, a user of the parameterized score estimation software tool **302** may thus iteratively or otherwise configurably change input parameters used by the parameterized score estimation software tool **302**, such as until a desired score estimation is output. The desired score estimation may thus represent a desired design expression for the current project which maximizes an expected cost-benefit ratio thereof. In some implementations, the specific values of the input parameters used to achieve the desired score estimation may then be recorded, such as within the database of the automated parameterized modeling and scoring intelligence platform **300**.

(49) The parameterized score optimization software tool **304** is a software tool which processes a set of inputs using external data to output a score optimization. The parameterized score optimization software tool **304** extrapolates upon the set of inputs according to external application data to determine the score optimization. In the example in which the parameterized score optimization software tool **304** is used for estimating scores for construction projects, the set of inputs may be or otherwise refer to a set of design metrics representing first input parameter values and a set of historical project data representing second input parameter values, and the external application data may be or otherwise refer to real-world property information, such as market value attributions, representing third input parameter values.

(50) In some implementations, the extrapolation of the set of inputs may be based on processing performed by or otherwise using the parameterized score estimation software tool **302**. For example, the first input parameter values and/or the second input parameter values, as the set of design metrics and the historical project data, respectively, may be parameter values used by or otherwise processed using the parameterized score estimation software tool **302**. In some

implementations, the first input parameter values and/or the second input parameter values may be recorded by the parameterized score estimation software tool **302**, such as within a database (e.g., the internal database **208**), and thereafter retrieved by the parameterized score optimization software tool **304**. In some implementations, the first input parameter values and/or the second input parameter values may be received as input at the parameterized score optimization software tool **304**, such as from a user thereof.

(51) The real-world property information may include or otherwise refer to information representing property values for one or more real property sites. The real-world property information, as external application data, may be retrieved from one or more sources. For example, an external application from which the external application data may be retrieved include government websites (e.g., Register of Deeds or other department offices), commercial property assessment web sites which track real-world property information, or other sources, which may be online or offline.

(52) The real-world property information may represent a current market value for a real property site. The current market value may be processed by the parameterized score optimization software tool **304** in connection with the set of design metrics and the historical project data to determine score optimizations. For example, a score optimization may refer to an optimized market valuation for a current project. A score optimization may be determined by processing the set of design metrics, representing design considerations for the current project, according to the historical project data, representing guides for resource estimation, and according to the current market value of one or more real property sites, representing an expected yield based on specific design qualities of those one or more real property sites. The one or more real property sites used may be selected by a user of the parameterized score optimization software tool **304**. Alternatively, the parameterized score optimization software tool **304** may select the one or more real property sites, such as using an external application (e.g., the external application **212**) and such as based on project similarities between the current project and such real property sites as may be available.

(53) The score optimization determined and output by the parameterized score optimization software tool **304** may include or indicate an expected yield for the current project based on the set of design metrics, the historical project data, and the current market value of the one or more real property sites. The expected yield may represent or otherwise indicate an expected return on one or more resources to use to complete the current project.

(54) After a first score optimization is output, a user of the parameterized score optimization software tool **304** may modify one or more values of the design metrics, enable or disable one or more design metric categories, change the selection of historical projects used, and/or change the selection of real property sites used. After making such changes, the user may cause the parameterized score optimization software tool **304** to determine a new score based on the changes. The score optimization output after the changes are made may not be the same as the score optimization output before the changes are made. In some implementations, a user of the parameterized score optimization software tool **304** may thus iteratively or otherwise configurably change input parameters used by the parameterized score optimization software tool **304**, such as until a desired score optimization is output. The desired score optimization may thus represent a desired design expression for the current project which maximizes the expected yield thereof. In some implementations, the specific values of the input parameters used to achieve the desired score optimization may then be recorded, such as within the database of the automated parameterized modeling and scoring intelligence platform **300**.

(55) The web crawler deployment software tool **306** deploys web crawlers to a network to retrieve data which may be used by one or both of the parameterized score estimation software tool **302** or the parameterized score optimization software tool **304**. A web crawler deployed by the web crawler deployment software tool **306** is a network bot configured to search through a data source (e.g., the internet). The web crawler may be configured for data scraping of one or more data

sources. For example, the web crawler may be configured to search for data using an internet search engine, download some or all of a webpage, and extract some or all information from the downloaded webpage. The web crawler may be configured to report when data is retrieved, such as on an individual basis or on a batch basis (e.g., after a number of data pieces are retrieved). The data retrieved using a web crawler may be stored in a database of the automated parameterized modeling and scoring intelligence platform **300** (e.g., the internal database **208**). A web crawler deployed by the web crawler deployment software tool **306** may use one or more techniques for crawling, including, without limitation, text pattern matching, HTTP programming, HTML parsing, DOM parsing, vertical aggregation, semantic annotation recognition, screen scraping, or computer vision analysis.

(56) The application and database interfaces **308** are software aspects used to interface portions of the automated parameterized modeling and scoring intelligence platform **300** with external applications (e.g., the external application **212** shown in FIG. **2**), databases (e.g., the internal database **208**), and/or other data sources (e.g., Internet services which may be used to locate, identify, or otherwise retrieve data from an external database, such as using an external application). The application and database interfaces **308** may be or otherwise refer to a set of APIs, database management system interfaces, or the like. The application and database interfaces **308** may be included as default aspects of the automated parameterized modeling and scoring intelligence platform **300**. In some implementations, the application and database interfaces **308** may be extended or otherwise configured, such as by a user of the automated parameterized modeling and scoring intelligence platform **300**.

(57) The GUIs **310** are GUIs which may be rendered or displayed, such as to render or display aspects of the parameterized score estimation software tool **302**, the parameterized score optimization software tool **304**, other functionality of the automated parameterized modeling and scoring intelligence platform **300**, or a combination thereof. A GUI of the GUIs **310** can comprise part of a software GUI constituting data that reflect information ultimately destined for display on a hardware device, for example, the client **104** shown in FIG. **1**). For example, the data can contain rendering instructions for bounded graphical display regions, such as windows, or pixel information representative of controls, such as buttons and drop-down menus. The rendering instructions can, for example, be in the form of HTML, SGML, JavaScript, Jelly, AngularJS, or other text or binary instructions for generating the GUI or another GUI on a display that can be used to generate pixel information. A structured data output of one device can be provided to an input of the hardware display so that the elements provided on the hardware display screen represent the underlying structure of the output data.

(58) The machine learning models **312** are software aspects which may be trained according to various input, such as those associated with past requests made to the automated parameterized modeling and scoring intelligence platform **300** (e.g., the requests **202** shown in FIG. **2**) and which may thereafter be used for inference according to such training, for example, to identify, select, determine, or otherwise generate aspects associated with the processing of future requests made to the automated parameterized modeling and scoring intelligence platform **300**. The machine learning models **312** apply intelligence to identify complex patterns in the input and to leverage those patterns to produce output and refine systemic understanding of how to process the input to produce the output. The machine learning models **312** may be used by the parameterized score estimation software tool **302** and/or by the parameterized score optimization software tool **304**. The machine learning models **312** may be or include one or more of a neural network (e.g., a convolutional neural network, recurrent neural network, or other neural network), decision tree, vector machine, Bayesian network, genetic algorithm, deep learning system separate from a neural network, or other machine learning model.

(59) FIG. **4** is a block diagram of examples of functionality of a parameterized score estimation software tool **400** of an automated parameterized modeling and scoring intelligence platform, such

as the automated parameterized modeling and scoring intelligence platform **200** shown in FIG. **2**. For example, the parameterized score estimation software tool **400** may be the parameterized score estimation software tool **203** shown in FIG. **3**. The parameterized score estimation software tool **400** in the example as shown includes a parameter definition tool **402**, a parameter modeling tool **404**, a constraint processing tool **406**, and a project scoring tool **408**.

(60) The parameter definition tool **402** is used to define first input parameters used by the parameterized score estimation software tool **400**. The first input parameters are design metrics **410**. The design metrics are categories of current project design criteria which specify aspects of the current project design. The values provided as the design metrics **410** are used by the parameter definition tool **402** to define the first input parameters. Examples of the categories of design metrics to which the values provided as the design metrics **410** may correspond to include, but are not limited to, region, product type, construction type, lot size, construction duration, unit, average unit size, bedroom, bathroom, net rentable square footage, efficiency, gross square footage in wooden structure, wood levels, liner unit gross square footage, common area open square footage, pool/spa square footage, parking ratio, parking efficiency, garage levels, recreation/leasing, and retail. Units of measurement for some or all of these categories may vary.

(61) In some implementations, the design metrics **410** may be further categorized into general information, building information, amenity and retail information, and garage information. For example, the general information may include design metrics relating to region, gross lot size, onsite street square footage, product type, and/or construction type. In another example, the building information may include design metrics relating to unit count, unit types (e.g., size, bedrooms, bathrooms, etc.), efficiency, and/or units in concrete. In yet another example, the amenity and retail information may include design metrics relating to amenities in wood, amenities in concrete, standalone amenities, retail in wood, retail in concrete, and/or standalone retail. In yet a further example, the garage information may include design metrics relating to parking information, parking efficiency, and/or garage levels.

(62) Defining the first input parameters using the design metrics **410** includes plugging the values of the design metrics **410** into the respective parameter fields used by the parameterized score estimation software tool **400**. In some implementations, the design metrics **410** may be received from or otherwise indicated by a user of the parameterized score estimation software tool **400**. In some implementations, the parameter definition tool **402** may retrieve the design metrics **410**, such as from a database of the automated parameterized modeling and scoring intelligence platform.

(63) In some implementations, the design metrics **410** may be primary design metrics, and the parameter definition tool **402** can calculate secondary design metrics based on the primary design metrics. For example, categories of the design metrics **410** may be arranged hierarchically such that certain values within the design metrics **410** may feasibly fall within more than one primary design metric category, but only one secondary design metric category. The parameter definition tool **402** can calculate the secondary design metrics based on the context of the primary design metrics, for example, with reference to a value and unit of measurement therefore in connection with the specified primary design metric category.

(64) The design metrics **410** may not represent a full take off of a design for the current project, for example, by listing complete information for linear square footage, square footage, and the like for various values. This may be because the design process is still in an early stage. Instead, a design metric that is readily available at early design stages and has a high correlation coefficient for current project items (e.g., concrete, windows, stucco, grading, etc.) may be designated as an alternative unit of measure.

(65) The parameter modeling tool **404** is used to model the parameters defined using the parameter definition tool **402** according to historical project data **412**. The historical project data **412** includes data relevant for modeling the design metrics **410** in the context of the current project, such as by evaluating comparisons to historical projects. The historical project data **412** may correspond to

one or more historical projects, which may be selected by a user of the automated parameterized modeling and scoring intelligence platform or by the platform itself (e.g., by or otherwise using the parameter modeling tool **404** or another software tool). The historical projects used to identify the historical project data **412** may be selected based on one or more of a common budget scope with the current project, a common project type with the current project, a common set of materials or other design metrics with the current project, or other bases.

(66) In some implementations, the parameter modeling tool **404** may calculate a similarity measurement for a historical project before it is selected. For example, the similarity measurement may represent values resulting from comparing one or more aspects of the historical project to one or more corresponding aspects of the current project. In some implementations, a historical project may be further considered only if the similarity measurement meets a threshold, which may be defined by the parameter modeling tool **404** or by a user thereof. In some implementations, multiple historical projects may be used together only where the similarity measurements determined between those multiple historical projects meets a threshold, which may be defined by the parameter modeling tool **404** or by a user thereof. The similarity measurement can be determined based on a one-to-one comparison of relevant design metrics. Alternatively, the similarity measurement can be weighted based on data relevance. For example, inputs corresponding to region, product type, and construction type may be considered the most relevant for the purpose of determining the similarity measurement.

(67) Modeling the parameters defined using the parameter definition tool **402** can include identifying the parameters that correspond to values of the historical project data **412**. For example, the parameter modeling tool **404** can process the historical project data **412** to determine which aspects of the historical project data **412** to use to process which aspects of the design metrics **410**. In some implementations, the parameter modeling tool **404** may further model the defined parameters by applying weights to some or all of the defined parameters. For example, the weights may be defined by the historical project data **412**, such as where the historical project data **412** indicates a greater relevance or importance of certain types of the design metrics **410** over other types thereof.

(68) In some implementations, the parameterized score estimation software tool **400** may automate an analysis of a collection of historical project data to recommend certain project data as the historical project data **412** for a current project. For example, certain sets of historical project data may be considered more relevant and therefore selected for certain projects and/or for certain materials/services used for certain projects, such as may be indicated by the design metrics **410**.

(69) The historical project data **412** may represent data for previous projects of a user of the automated parameterized modeling and scoring intelligence platform who is using the parameterized score estimation software tool **400** for a current project. However, in some implementations, the parameterized score estimation software tool **400** may use historical project data from other users of the automated parameterized modeling and scoring intelligence platform. For example, where the automated parameterized modeling and scoring intelligence platform is implemented in a multi-tenant computing environment in which different users provide different sets of historical project data into a shared library, and the parameterized score estimation software tool **400** may select the historical project data **412** from the shared library. The historical project data may be anonymized such that individual historical project data items are not usable to identify the users from whom those data items derived.

(70) The constraint processing tool **406** processes the modeled parameters against constraints **414** to control limits and other restrictions against the current project. Examples of the constraints **414** may include, but are not limited to, zoning requirements, covenants, envelope requirements, unit maximums, height maximums, and the like. The constraints **414** may be defined by or otherwise received from a user of the parameterized score estimation software tool **400**. Alternatively, the constraints **414** may be retrieved from one or more external sources. For example, the constraints

414 may be retrieved from a website of a municipal zoning or like department office for the municipality in which the property associated with the current project is located. The constraint processing tool **406** uses the constraints **414** to cull design considerations which are not appropriate, such as because the constraint processing tool **406** recognizes them as violating one or more of the constraints **414**.

(71) The project scoring tool **408** receives the constraint-processed project information and determines a score estimation **416** for the current project based thereon. Score estimations can be determined for the current project as a whole and/or for individual items used in the current project. To estimate a score for the entire current project, the individual scores calculated for the various design metrics **410** modeled according to the historical project data **412** may be added, averaged, or input into a specialized formula which ultimately outputs the score estimation **416**. A score for a given one of the design metrics **410** may be determined by leveraging corresponding data of the historical project data **412**. For example, determining a score for a given one of the design metrics **410** may include calculating a cost escalation percentage for the historical project data, calculating the unit cost of each item based on the unit measurement preset therefor, and calculating the cost of each item by multiplying the calculated unit by the quantity associated with the unit of measure for the item.

(72) An example of determining a score for a design metric corresponding to windows for a current project is now described. First, the parametric score estimation software tool **400** receives the design metrics **410** corresponding to the total number of bedrooms in the current project and a selection of some number of historical projects to use information from as the historical project data **412**. The parametric score estimation software tool **400** retrieves information indicating a cost of windows and a total number of bedrooms from the selected historical projects. The parametric score estimation software tool **400** uses the retrieved information to calculate a unit cost representing a cost per bedroom count for the selected historical projects. The parametric score estimation software tool **400** then escalates the unit cost at a predefined annual percentage rate based on a completion date for the selected historical projects and based on a current date. The parametric score estimation software tool **400** then calculates an average of the escalated unit costs of the selected historical projects and multiples that calculated unit cost by the number of bedrooms specified in the design metrics **410** to determine the score estimation as a cost for the windows.

(73) The score estimation **416** may be used to iterate the design metrics **410** and/or the historical project data **412**, such as to result in a determination of a different score estimation. For example, a user of the parameterized score estimation software tool **400** can, based on the score estimation **416**, change one or more aspects of the design metrics **410** and/or of the historical project data **412** to cause the parameterized score estimation software tool **400** to produce a different score estimation. In another example, the automated parameterized modeling and scoring intelligence platform which includes the parameterized score estimation software tool **400** may automate an iteration of the parameterized score estimation software tool **400** by changing one or more aspects of the design metrics **410** and/or of the historical project data **412** to cause the parameterized score estimation software tool **400** to produce a different score estimation. In some implementations, a design associated with the score estimation **416** may be transmitted to a user of the automated parameterized modeling and scoring intelligence platform. For example, the design associated with the score estimation **416** may include a list of items or materials to use in the current project. In another example, the design associated with the score estimation **416** may include a visual render of the design for the current project.

(74) In some implementations, the parameterized score estimation software tool **400** can use machine learning functionality of the automated parameterized modeling and scoring intelligence platform, or such machine learning functionality can otherwise be used, to determine the score estimation **416**. For example, a machine learning model (e.g., of the machine learning models **312** shown in FIG. **3**) may be trained using training data sets made up on data samples. The data

samples may include sets of training data including one or more pre-processed data points and one or more corresponding post-processed data points. For example, the data samples may include an initial parameter value (e.g., of the design metrics **410** or of the historical project data **412**), an initial score estimation (e.g., the score estimation **416** before changes are made to the design metrics **410** or to the historical project data **412**), a changed parameter value (e.g., the same parameter of the design metrics **410** or of the historical project data **412**), and a new score estimation (e.g., the score estimation **416** resulting from the changes made to the design metrics **410** or to the historical project data **412**). The data samples are used to train the machine learning model to recognize patterns in the behavior and/or relationships of the data. The training may include binary and/or multiclass classification, which may be supervised or unsupervised. In some embodiments, the classification can be performed using Naïve Bayes, K means clustering, or another approach.

(75) In another example, a trained machine learning model may be used to perform inference against data sets, which may include the design metrics **410**, the historical project data **412**, and/or the score estimation **416**. Performing inference against a data set includes applying patterns recognized by the training to the current data set to determine how to change some or all of the data, such as to arrive at a maximized or otherwise improved score estimation. For example, performing inference against a data set may include determining to change certain values of the design metrics **410** because the machine learning model training recognizes that the ultimate value gained from having a certain amount of some type of item or material is less than the ultimate cost associated therewith.

(76) In some implementations, the parameter modeling tool **404** can output an alert indicating a scale warning when comparisons made between the design metrics **410** and values of the historical project data **412** are not identical or similar. For example, an alert may be output where a current project is for the development of a ten acre land size whereas a historical project instead was for the development of a one acre land size. The scale warning can thus be used to alert the software and/or the user that the score to be estimated for the current project will be substantively different from a score which may have been estimated for the corresponding historical project. For example, excavation costs may be higher even though the land purchase cost will be lower. In some implementations, machine learning or other intelligence functionality of the automated parameterized modeling and scoring intelligence platform can analyze differences in the project scales based on the outputs of scale warnings to identify scaling patterns and to apply those patterns upon the outputting of an alert.

(77) FIG. **5** is a block diagram of examples of functionality of a parameterized score optimization software tool **500** of an automated parameterized modeling and scoring intelligence platform, such as the automated parameterized modeling and scoring intelligence platform **200** shown in FIG. **2**. For example, the parameterized score optimization software tool **500** may be the parameterized score optimization software tool **304** shown in FIG. **3**. The parameterized score optimization software tool **500** in the example as shown includes a score estimation tool **502**, a regression processing tool **504**, a pro forma generation tool **506**, and an expected yield candidate selection tool **508**.

(78) The score estimation tool **502** processes design metrics **510** and historical project data **512** to determine a score estimation for a current project based on a current design thereof. The design metrics **510** and the historical project data **512** may respectively be or otherwise refer to the design metrics **410** and the historical project data **412** shown in FIG. **4** after iteration to arrive at a desired score estimation. In some implementations, the score estimation tool **502** may be or refer to a parameterized score estimation software tool, for example, the parameterized score estimation software tool **400** shown in FIG. **4**. The output of the score estimation tool **502** is a score estimation, which may, for example, represent a number of resources expected to be used for the current project.

(79) The regression processing tool **504** processes external application data **514** against the output of the score estimation tool **502** to estimate a market value for the current project based on the current design thereof. The external application data **514** may be or otherwise refer to information associated with an external application, such as data stored within the external database **214** of the external application **212** shown in FIG. **2**. For example, the external application data **514** may be or otherwise refer to market value information indicating values of one or more real property sites. In some implementations, the external application data **514** may be received directly from an external application, such as by the regression processing tool **504** or another aspect of the automated parameterized modeling and scoring intelligence platform making an API call to the external application. In some implementations, the external application data **514** may be retrieved using web crawlers deployed to the external application.
(80) The regression processing tool **504** performs a regression analysis against the score estimation based on the external application data **514** to determine a market value estimate (e.g., as a rent estimate or other estimate) for aspects of one or more different projects (e.g., based on location, unit types (e.g., studio, one bedroom, etc.), unit sizes, etc.). For example, the regression processing tool **504** can use the external application data **514** to analyze and identify market value estimates for some or all of the different project aspects. Those market value estimates may then be used to determine the output of the regression processing tool **504**. The output of the score estimation tool **502** may indicate score estimations for individual ones of the design metrics **510**. The regression processing tool **504** can perform a regression analysis against combinations of those score estimations or against individual ones of those score estimations. The output of the regression processing tool **504** is a set of data indicating relationships between aspects of the current project and a market value estimate for the current project.
(81) The pro forma generation tool **506** generates a pro forma for the current project based on the output of the regression processing tool **504**. The pro forma generated by or otherwise using the pro forma generation tool **506** includes a set of records detailing some or all of the different construction projects for which market value estimates were determined by or otherwise using the regression processing tool **504** along with some or all of the score estimations determined by or otherwise using the score estimation tool **502**. The pro forma may, for some or all of the different projects, indicate expected yields for those projects.
(82) The expected yield candidate selection tool **508** provides expected yield candidates for selection, such as by a user of the automated parameterized modeling and scoring intelligence platform and/or by the platform itself. The expected yield candidate selection tool **508** processes feedback **516** against the output of the pro forma generation tool **506** to determine, generate, produce, or otherwise identify a set of expected yield candidates. The feedback **516** may be provided by a user of the automated parameterized modeling and scoring intelligence platform and/or by the platform itself. The feedback **516** may indicate to cull certain expected yield candidates, to temporarily ignore certain expected yield candidates, to further consider certain expected yield candidates, and/or to select an expected yield candidate as an expected yield **518** for ultimate output from the parameterized score optimization software tool **500**.
(83) Prior to selecting one of the expected yield candidates as the expected yield **518**, a user of the parameterized score optimization software tool **500** may iterate against the output of the expected yield candidate selection tool **508**, such as by changing one or more aspects of the design metrics **510**, the historical project data **512**, and/or the external application data **514**. For example, where the design metrics **510** are expressed hierarchically, a user of the parameterized score optimization software tool **500** may change a primary design metric, which may cause or otherwise implicate the changing of one or more secondary design metrics. In some implementations, the parameterized score optimization software tool **500** itself may iterate against the output of the expected yield candidate selection tool **508**, such as based on a set of constraints for the expected yield to select as the output of the parameterized score optimization software tool **500**. For example, a constraint

used to iterate against the output of the expected yield candidate selection tool **508** may indicate a minimum expected yield value. In the event the user of the parameterized score optimization software tool **500** or the software tool itself identifies a desired expected yield candidate, that candidate is output from the parameterized score optimization software tool **500** as the expected yield **518**. In some implementations, a design associated with the expected yield **518** may be transmitted to a user of the automated parameterized modeling and scoring intelligence platform. For example, the design associated with the expected yield **518** may include a list of items or materials to use in the current project. In another example, the design associated with the expected yield **518** may include a visual render of the design for the current project.

(84) The parameterized score optimization software tool **500** may thus be used to iterate through different design candidates for a current project and to identify an optimal one of those designs as the design which has the highest expected yield of the different design candidates. The automated expected yield candidate determination by the parameterized score optimization software tool **500** may thus be used to maximize a yield for the current project, such as by enabling a user thereof to identify design aspects which maximize the yield for his or her current project.

(85) In some implementations, the parameterized score optimization software tool **500** may include or otherwise use a recommendation engine to recommend an expected yield candidate as the expected yield **518** output from the parameterized score optimization software tool **500**. For example, the recommendation engine may analyze the expected yield candidates to determine values of the expected yields for each of them. The recommendation engine may then rank the expected yield candidates according to the determined values. In some implementations, the N (e.g., five) highest ranked expected yield candidates may be output for user review. The value of N may be configurable by a user or set by default. In some implementations, the recommendation engine may automatically output the highest ranked expected yield candidate as the expected yield **518** output from the parameterized score optimization software tool **500**.

(86) In some implementations, the parameterized score optimization software tool **500** can use machine learning functionality of the automated parameterized modeling and scoring intelligence platform, or such machine learning functionality can otherwise be used, to determine the expected yield **518**. For example, a machine learning model (e.g., of the machine learning models **312** shown in FIG. **3**) may be trained using training data sets made up on data samples. The data samples may include sets of training data including one or more pre-processed data points and one or more corresponding post-processed data points. For example, the data samples may include an initial parameter value (e.g., of the design metrics **510**, of the historical project data **512**, or of the external application data **514**), an initial expected yield (e.g., the expected yield **518** before changes are made to the design metrics **510**, to the historical project data **512**, or to the external application data **514**), a changed parameter value (e.g., the same parameter of the design metrics **510**, of the historical project data **512**, or of the external application data **514**), and a new expected yield (e.g., the expected yield **518** after changes are made to the design metrics **510**, to the historical project data **512**, or to the external application data **514**). The data samples are used to train the machine learning model to recognize patterns in the behavior and/or relationships of the data. The training may include binary and/or multiclass classification, which may be supervised or unsupervised. In some embodiments, the classification can be performed using Naïve Bayes, K means clustering, or another approach.

(87) In another example, a trained machine learning model may be used to perform inference against data sets, which may include the design metrics **510**, the historical project data **512**, the external application data **514**, and/or the expected yield **518**. Performing inference against a data set includes applying patterns recognized by the training to the current data set to determine how to change some or all of the data, such as to arrive at a maximized or otherwise improved expected yield. For example, performing inference against a data set may include determining to change certain values of the design metrics **510** the machine learning model training recognizes that the

ultimate expected yield gained from having a certain amount of some type of item or material is less than the ultimate cost associated therewith.

(88) FIG. **6** is an illustration showing an example of a GUI **600** of a parameterized score estimation software tool, such as the parameterized score estimation software tool **400** shown in FIG. **4**. For example, the GUI **600** may be one of the GUIs **310** described above with respect to FIG. **3**. The example of the GUI **600** as shown includes a design section on the left, a comps section in the middle, and a score section on the right. The design section displays information associated with the design metrics and values therefor which are defined as parameters used by the parameterized score estimation software tool. The comps section displays information associated with one or more selected historical projects, including measures of similarity to the current project and other relevant comparisons between the selected historical projects. The score section displays aspects of the score estimation determined and output by the parameterized score estimation software tool.

(89) FIG. **7** is an illustration showing an example of a GUI **700** of a parameterized score optimization software tool, such as the parameterized score optimization software tool **500** shown in FIG. **5**. For example, the GUI **700** may be one of the GUIs **310** described above with respect to FIG. **3**. In particular, the GUI **700** shows an example of output of the parameterized score optimization software tool in the form of a chart. In some implementations, the GUI **700** may include a table in addition to or instead of the chart. The paired bars represent incremental design changes to the current project that creates a new design. In particular, the leftmost bar of each pair may represent a change in base points compared to the original project design, and the rightmost bar of each pair may represent a change in the overall net rentable square footage compared to the original project design.

(90) To further describe some implementations in greater detail, reference is next made to examples of techniques which may be performed by or using a region-based electrical intelligence system as described with respect to FIGS. **1-7**. FIG. **8** is a flowchart showing an example of a technique **800** for parameterized score estimation using an automated parameterized modeling and scoring intelligence system. FIG. **9** is a flowchart showing an example of a technique **900** for parameterized score optimization using an automated parameterized modeling and scoring intelligence system.

(91) The technique **800** and/or the technique **900** can be executed using computing devices, such as the systems, hardware, and software described with respect to FIGS. **1-7**. The technique **800** and/or the technique **900** can be performed, for example, by executing a machine-readable program or other computer-executable instructions, such as routines, instructions, programs, or other code. The steps, or operations, of the technique **800** and/or the technique **900** or another technique, method, process, or algorithm described in connection with the implementations disclosed herein can be implemented directly in hardware, firmware, software executed by hardware, circuitry, or a combination thereof.

(92) For simplicity of explanation, the technique **800** and the technique **900** are both depicted and described herein as a series of steps or operations. However, the steps or operations in accordance with this disclosure can occur in various orders and/or concurrently. Additionally, other steps or operations not presented and described herein may be used. Furthermore, not all illustrated steps or operations may be required to implement a technique in accordance with the disclosed subject matter.

(93) Referring first to FIG. **8**, the technique **800** for parameterized score estimation using an automated parameterized modeling and scoring intelligence system is shown. At **802**, design metrics for a current project are identified. At **804**, parameters for score estimation using the design metrics are defined. At **806**, one or more historical projects to use for modeling are selected. At **808**, the defined parameters are modeled using the historical project data. At **810**, the modeled parameters are processed against constraints. At **812**, a score estimation is determined for the processed parameters. At **814**, one or more of the initial parameters are changed to iterate the score

estimation. At **816**, a new score estimation is determined based on the changed parameter aspects. In some implementations, the technique **800** may include outputting a design associated with the new score estimation, such as by transmitting the design to a client device from which input parameters are received.

(94) Referring next to FIG. **9**, the technique **900** for parameterized score optimization using an automated parameterized modeling and scoring intelligence system is shown. At **902**, a score estimation is determined for the current project based on design metrics for a current project and according to historical project data corresponding to one or more historical projects selected for modeling the design metrics. At **904**, a regression analysis is performed against the score estimation based on external application data to determine value estimate data for the current project. At **906**, a pro forma is generated based on the value estimate data. At **908**, feedback indicating further processing of the pro forma is received. At **910**, expected yield candidates are identified based on the feedback. At **912**, one or more parameters are changed to iterate the score optimization. At **914**, new expected yield candidates are identified based on the changed parameters. At **916**, an expected yield candidate is selected from the new expected yield candidates. In some implementations, the technique **900** may include outputting a design associated with the selected expected yield candidate, such as by transmitting the design to a client device from which input parameters are received.

(95) FIG. **10** is a block diagram showing an example of a computing device **1000** which may be used in an automated parameterized modeling and scoring intelligence system, for example, the automated parameterized modeling and scoring intelligence system **100** shown in FIG. **1**. The computing device **1000** may be used to implement a server which implements some or all of a service environment (e.g., the service environment **108** shown in FIG. **1**). Alternatively, the computing device **1000** may be used to implement a client that accesses the server (e.g., the client **104** shown in FIG. **1**). As a further alternative, the computing device **1000** may be used as or to implement another client, server, or other device according to the implementations disclosed herein.

(96) The computing device **1000** includes components or units, such as a processor **1002**, a memory **1004**, a bus **1006**, a power source **1008**, peripherals **1010**, a user interface **1012**, and a network interface **1014**. One of more of the memory **1004**, the power source **1008**, the peripherals **1010**, the user interface **1012**, or the network interface **1014** can communicate with the processor **1002** via the bus **1006**.

(97) The processor **1002** is a central processing unit, such as a microprocessor, and can include single or multiple processors having single or multiple processing cores. Alternatively, the processor **1002** can include another type of device, or multiple devices, now existing or hereafter developed, configured for manipulating or processing information. For example, the processor **1002** can include multiple processors interconnected in any manner, including hardwired or networked, including wirelessly networked. For example, the operations of the processor **1002** can be distributed across multiple devices or units that can be coupled directly or across a local area or other suitable type of network. The processor **1002** can include a cache, or cache memory, for local storage of operating data or instructions.

(98) The memory **1004** includes one or more memory components, which may each be volatile memory or non-volatile memory. For example, the volatile memory of the memory **1004** can be random access memory (RAM) (e.g., a DRAM module, such as DDR SDRAM) or another form of volatile memory. In another example, the non-volatile memory of the memory **1004** can be a disk drive, a solid state drive, flash memory, phase-change memory, or another form of non-volatile memory configured for persistent electronic information storage. The memory **1004** may also include other types of devices, now existing or hereafter developed, configured for storing data or instructions for processing by the processor **1002**.

(99) The memory **1004** can include data for immediate access by the processor **1002**. For example,

the memory **1004** can include executable instructions **1016**, application data **1018**, and an operating system **1020**. The executable instructions **1016** can include one or more application programs, which can be loaded or copied, in whole or in part, from non-volatile memory to volatile memory to be executed by the processor **1002**. For example, the executable instructions **1016** can include instructions for performing some or all of the techniques of this disclosure. The application data **1018** can include user data, database data (e.g., database catalogs or dictionaries), or the like. The operating system **1020** can be, for example, Microsoft Windows®, Mac OS X®, or Linux®; an operating system for a small device, such as a smartphone or tablet device; or an operating system for a large device, such as a mainframe computer.

(100) The power source **1008** includes a source for providing power to the computing device **1000**. For example, the power source **1008** can be an interface to an external power distribution system. In another example, the power source **1008** can be a battery, such as where the computing device **1000** is a mobile device or is otherwise configured to operate independently of an external power distribution system.

(101) The peripherals **1010** includes one or more sensors, detectors, or other devices configured for monitoring the computing device **1000** or the environment around the computing device **1000**. For example, the peripherals **1010** can include a geolocation component, such as a global positioning system location unit. In another example, the peripherals can include a temperature sensor for measuring temperatures of components of the computing device **1000**, such as the processor **1002**.

(102) The user interface **1012** includes one or more input interfaces and/or output interfaces. An input interface may, for example, be a positional input device, such as a mouse, touchpad, touchscreen, or the like; a keyboard; or another suitable human or machine interface device. An output interface may, for example, be a display, such as a liquid crystal display, a cathode-ray tube, a light emitting diode display, or other suitable display.

(103) The network interface **1014** provides a connection or link to a network (e.g., the network **116** shown in FIG. **1**). The network interface **1014** can be a wired network interface or a wireless network interface. The computing device **1000** can communicate with other devices via the network interface **1014** using one or more network protocols, such as using Ethernet, TCP, IP, power line communication, Wi-Fi, Bluetooth, infrared, GPRS, GSM, CDMA, Z-Wave, ZigBee, another protocol, or a combination thereof.

(104) Implementations of the computing device **1000** may differ from what is shown and described with respect to FIG. **10**. In some implementations, the computing device **1000** can omit the peripherals **1010**. In some implementations, the memory **1004** can be distributed across multiple devices. For example, the memory **1004** can include network-based memory or memory in multiple clients or servers performing the operations of those multiple devices. In some implementations, the application data **1018** can include functional programs, such as a web browser, a web server, a database server, another program, or a combination thereof.

(105) The implementations of this disclosure can be described in terms of functional block components and various processing operations. Such functional block components can be realized by a number of hardware or software components that perform the specified functions. For example, the disclosed implementations can employ various integrated circuit components (e.g., memory elements, processing elements, logic elements, look-up tables, and the like), which can carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, where the elements of the disclosed implementations are implemented using software programming or software elements, the systems and techniques can be implemented with a programming or scripting language, such as C, C++, Java, JavaScript, assembler, or the like, with the various algorithms being implemented with a combination of data structures, objects, processes, routines, or other programming elements.

(106) Functional aspects can be implemented in algorithms that execute on one or more processors. Furthermore, the implementations of the systems and techniques disclosed herein could employ a

number of conventional techniques for electronics configuration, signal processing or control, data processing, and the like. The words “mechanism” and “component” are used broadly and are not limited to mechanical or physical implementations, but can include software routines in conjunction with processors, etc.
(107) Likewise, the terms “system” or “mechanism” as used herein and in the figures, but in any event based on their context, may be understood as corresponding to a functional unit implemented using software, hardware (e.g., an integrated circuit, such as an ASIC), or a combination of software and hardware. In certain contexts, such systems or mechanisms may be understood to be a processor-implemented software system or processor-implemented software mechanism that is part of or callable by an executable program, which may itself be wholly or partly composed of such linked systems or mechanisms.
(108) Implementations or portions of implementations of the above disclosure can take the form of a computer program product accessible from, for example, a computer-usable or computer-readable medium. A computer-usable or computer-readable medium can be any device that can, for example, tangibly contain, store, communicate, or transport a program or data structure for use by or in connection with any processor. The medium can be, for example, an electronic, magnetic, optical, electromagnetic, or semiconductor device.
(109) Other suitable mediums are also available. Such computer-usable or computer-readable media can be referred to as non-transitory memory or media, and can include volatile memory or non-volatile memory that can change over time. A memory of an apparatus described herein, unless otherwise specified, does not have to be physically contained by the apparatus, but is one that can be accessed remotely by the apparatus, and does not have to be contiguous with other memory that might be physically contained by the apparatus.
(110) While the disclosure has been described in connection with certain implementations, it is to be understood that the disclosure is not to be limited to the disclosed implementations but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Claims

1. A system for automated parameterized modeling and scoring intelligence, the system comprising: a server device included in a service environment, the server device including a memory storing instructions, a network interface configured to provide communications between the server device and a client device, and a processor configured to execute the instructions stored in the memory to: train a machine learning model to determine an expected yield, wherein machine learning model is trained using one or more sets of historical project data, the historical project data including initial parameter values and initial expected yields for the initial parameter values; receive a data set for a current project, the data set including project parameters and historical project data stored within a database; implement the machine learning model to generate a first set of expected yields for a current project by determining a score estimation for the current project and performing a regression analysis against the score estimation based on application data associated with real property site information corresponding to the current project; implement the machine learning model to change at least some of the project parameters in the data set, to receive changed project parameters, by iterating through information associated with the current project based on the first set of expected yields according to patterns recognized from training data samples in the one or more sets of the historical project data; train again the machine learning model using the changed project parameters and the first set of expected yields; implement the machine learning model to determine a second set of expected yields for the current project based on second project parameters, the application data, and different values of the second project

parameters; and output project parameters corresponding to a selected expected yield from among the second set of expected yields, wherein the changed project parameters are used in generating a project design for the current project.
2. The system of claim 1, wherein the instructions include instructions to: determine a score estimation for a project using a set of project parameters and the historical project data; determine score estimations for different values of the project using the different values of the set of project parameters and the historical project data; determine a market value estimate for a project parameter based on the score estimation for the project and the score estimations for alternative configurations of the project; and output data indicating a relation between the project parameter the market value estimate of the project.
3. The system of claim 2, wherein the instructions include instructions to: perform a regression analysis on the first set of expected yields versus the different values of the set of project parameters to determine the market value estimate for a project parameter.
4. The system of claim 2, wherein the instructions include instructions to: generate a pro forma for the current project based on the relation between the project parameter and the market value estimate of the project.
5. The system of claim 1, wherein the instructions include instructions to: compare the set of project parameters against the one or more sets of historical data stored within the database to determine that a similarity measurement between the current project and one or more historical projects meets a threshold.
6. The system of claim 1, wherein the instructions include instructions to: deploy web crawlers to one or more data sources over a network, wherein the web crawlers are configured to retrieve the application data from the one or more data sources; and store the application data within the database.
7. The system of claim 1, wherein the instructions include instructions to: use a recommendation engine to rank different values of the project parameters or the different values of the second project parameters.
8. The system of claim 1, wherein the instructions include instructions to: transmit, using the network interface, rendering instructions to the client device for rendering one or more graphical user interfaces of the system for automated parameterized modeling and scoring intelligence at the client device.
9. The system of claim 1, wherein the project parameters include primary design metrics and secondary design metrics hierarchically arranged, wherein changes to one of the primary design metrics causes changes to a corresponding one or more of the secondary design metrics.
10. A method for automated parameterized modeling and scoring intelligence, the method comprising: training a machine learning model using one or more sets of historical project data to determine an expected yield for a given set of project parameter values, the historical project data including initial parameter values and initial expected yields for the initial parameter values; receiving a data set for a current project, the data set including project parameters and historical project data stored; implementing the machine learning model to generate a first set of expected yields for a current project by determining a score estimation for the current project and performing a regression analysis against the score estimation based on application data associated with real property site information corresponding to the current project; implementing the machine learning model to change at least some of the project parameters in the data set, to receive changed project parameters, by iterating through information associated with the current project based on the first set of expected yields according to patterns recognized from training data samples in the one or more sets of the historical project data; training again the machine learning model using the changed project parameters and the first set of expected yields; implementing the machine learning model to determine a second set of expected yields for the current project based on second project parameters, the application data, and different values of the second project parameters; and

outputting project parameters corresponding to a selected expected yield from among the second set of expected yields, wherein the changed project parameters are used in generating a project design for the current project.

11. The method of claim 10, further comprising: determining a score estimation for a project using a set of project parameters and the historical project data; determining score estimations for different values of the project using the different values of the set of project parameters and the historical project data; determining a market value estimate for a project parameter based on the score estimation for the project and the score estimations for alternative configurations of the project; and outputting data indicating a relation between the project parameter and the market value estimate of the project.

12. The method of claim 11, further comprising: performing a regression analysis on the first set of expected yields versus the different values of the set of project parameters to determine a market value estimate for a project parameter.

13. The method of claim 11, further comprising: generating a pro forma for the current project based on the relation between the project parameter and the market value estimate of the project.

14. The method of claim 10, further comprising: comparing the set of project parameters against the one or more sets of historical data stored within a database to determine that a similarity measurement between the current project and one or more historical projects meets a threshold.

15. The method of claim 10, further comprising: deploying web crawlers to one or more data sources over a network, wherein the web crawlers are configured to retrieve the application data from the one or more data sources; and storing the application data within a database.

16. The method of claim 10, further comprising: using a recommendation engine to rank different values of the project parameters and the different values of the second project parameters.

17. The method of claim 10, further comprising: transmitting rendering instructions to a client device for rendering one or more graphical user interfaces of a system for automated parameterized modeling and scoring intelligence.

18. A system for automated parameterized modeling and scoring intelligence, the system comprising: a server device including a processor and a memory, wherein the processor executes instructions stored in the memory to: train a machine learning model using one or more sets of historical project data to determine an expected yield for a given set of project parameter values, the historical project data including initial parameter values and initial expected yields for the initial parameter values; receive a data set for a current project, the data set including project parameters and historical project data stored within a database; implement the machine learning model to generate a first set of expected yields for a current project by determining a score estimation for the current project and performing a regression analysis against the score estimation based on application data associated with real property site information corresponding to the current project; the machine learning model to change at least some of the project parameters in the data set, to receive changed project parameters, by iterating through information associated with the current project based on the first set of expected yields according to patterns recognized from training data samples in the one or more sets of the historical project data; train again the machine learning model using the changed project parameters and the first set of expected yields; implement the machine learning model to determine a second set of expected yields for the current project based on second project parameters, the application data, and different values of the second project parameters; and output project parameters corresponding to a selected expected yield from among the second set of expected yields, wherein the changed project parameters are used in generating a project design for the current project.

19. The system of claim 18, wherein the instructions include instructions to: determine a score estimation for a project using a set of project parameters and historical project data; determine score estimations for different values of the project using the different values of the first set of project parameters and the historical project data; determine a market value estimate for a project

parameter based on the score estimation for the project and the score estimations for alternative configuration of the project; and output data indicating a relation between the project parameter the market value estimate of the project.

20. The system of claim 19, wherein the instructions include instructions to: perform a regression analysis on the first set of expected yields versus the different values of the set of project parameters to determine a market value estimate for a project parameter.